Cyclic quadrilaterals

1 $m = 106^\circ$

2

a 180°

b No, because opposite angles are equal not supplementary.

c Yes, because opposite angles are supplementary.

3 $d = 54^\circ$

4

a $y = 36$ (Opposite angles in a cyclic quadrilateral are supplementary)

b $p = 79$ (An exterior angle of a cyclic quadrilateral is equal to the angle that is interior and opposite to it)

c $x = 69$ (Angles forming a straight line are supplementary (sum to 180°))

d $n = 87$ (An exterior angle of a cyclic quadrilateral is equal to the angle that is interior and opposite to it)

5 $y = 77$

6

a $x = 108^\circ$

b $y = 70^\circ$

7

a $\angle AOC = 186^\circ$

b $\angle ABC = 93^\circ$

c No, because oppositice angles are not supplementary so it is not a cyclic quadrilateral.

8

a $m = 82^\circ$ (Angle at the centre of a circle is twice the angle at its circumference)

b $n = 82^\circ$

9

a $p = 90^\circ$ (The angle in a semicircle is a right angle)

b $q = 63^\circ$ (The interior angle sum of a triangle is 180°)

c $r = 117^\circ$ (Opposite angles in a cyclic quadrilateral are supplementary)

10

a $p = 112^\circ$ The exterior angle of a cyclic quadrilateral is equal to the opposite interior angle.

b $q = 68^\circ$ (Opposite angles in a cyclic quadrilateral are supplementary)

11 $m = 97^\circ$(m is the opposite exterior angle to $\angle DAB$)

12 $\angle BAD = 98^\circ$

13 $m = 69$(Opposite angles in a cyclic quadrilateral are supplementary)

14 $\angle BAF = 86$

Proofs

15 No solution provided.

16 No solution provided.

17 No solution provided.

18 No solution provided.

19 No solution provided.

20

a No solution provided.

b No solution provided.

c No solution provided.